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Hiring Team
Data and AI Engineering
The Brattle Group
Boston, Massachusetts

Dear Hiring Team,

I am applying for the Data and AI Engineer role in your Boston office. I completed an M.S. in Data Analytics for Science at Carnegie Mellon University in May 2026, with a 3.97 GPA, and a B.S. in Computer Science at the University of California, Santa Cruz. I am looking for an early-career role that stays close to imperfect data, asks for results a colleague can check, and builds technical judgment through real project work.

At Ashe System, I am founder and technical lead. I built a Python pipeline that scores aircraft-on-ground risk from operators' maintenance, inventory, and supplier exports. Those files were created for day-to-day operations, so the work started with inspection, cleaning, and a clear statement of what a score could support. Each recommendation keeps the evidence behind it, and the product is decision support for human review. Using XGBoost for temporal risk ranking, I improved validated Precision@20 from 35% to 62%. Tuning parameters and decision thresholds with Optuna increased median warning lead time from 6.5 to 10.2 days.

I have been applying the same standard to AI-assisted analysis. I co-founded AMOS and co-developed a Rust prototype that enforces data-access permissions and runs SQL and statistical calculations outside the language model. Schema and metric checks, evidence-linked claims, human review, and replay keep a result traceable after it is produced. I also built TraceFrame, local-first Python tooling for pandas, Polars, and DuckDB workflows. It records datasets, transformations, SQL outputs, quality checks, claims, and source-row lineage, and it exports an audit report. I use it so notebook and script results can be reviewed while the underlying data stays on the machine where the analysis ran.

In a Carnegie Mellon collaboration with PNC, I compared histogram gradient boosting, LightGBM, XGBoost, and an ensemble for ranking next-month product adoption. The pipeline reached 0.811 ROC AUC, and a supported recommendation appeared in the top three 95% of the time. We used those metrics because an advisor needs a short list that a person can review. As a software engineer intern at Wintec Industries, I built scheduled Python jobs for file ingestion, metadata indexing, and lifecycle tracking, and I wrote architecture and workflow documentation that improved onboarding and platform adoption by 35%.

Python and SQL are the tools I use for this work, along with Git, notebooks, and tests. I use AI-assisted development to learn and to move faster, and I verify generated output before I treat it as correct. I would like to bring that approach to the Data and AI Engineering team in Boston: preparing difficult source material, helping prototype methods that can become shared tools, and explaining assumptions, limitations, and tradeoffs to consultants and economists.

Thank you for considering my application. I would welcome a conversation.

Sincerely,

Elton Chang